Constructions of nonregular cospectral signed graphs

Callum Huntington

University of Naples Federico II, Italy

CSEECDMA 2024, Belgrade, Serbia

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Introduction

Some basics

A signed graph is a graph in which each edge is labelled as positive or negative. Formally a signed graph Σ is a pair $(G, \sigma) = G_\sigma$ where $G = (V, E)$ denotes an ordinary unsigned graph, called the underlying graph, with vertex set $V = \{v_1, v_2, \ldots, v_n\}$ and edge set $E = E(G)$ along with a map $\sigma : E \to \{+1, -1\}$.

The edge set of a signed graph is composed of subsets of positive and negative edges. Clearly the vertex sets $V(\Sigma)$ and $V(G)$ are identical and these may be used interchangeably but with the introduction of negative edges the same is generally not true for the edge sets $E(\Sigma)$ and $E(G)$.

We interpret an unsigned graph G as the all-positive signed graph $(G, +) = +G$ whose signature assigns the value $+1$ to all of its edges. Similarly by $(G, -) = -G$ we denote a graph G with the all-negative signature. In general for a signed graph $\Sigma = (G, \sigma)$ we have $-\Sigma = (G, -\sigma)$.

Adjacency and Laplacian matrices

The adjacency matrix A_Σ of $\Sigma = (G, \sigma)$ is obtained from the standard adjacency matrix of G by reversing the sign of all of the entries which correspond to negative edges. Formally the adjacency matrix of a signed graph $\Sigma = (G, \sigma)$ with n vertices is the $n \times n$ matrix denoted $A_\Sigma = (a_{ij})_{n\times n}$ and defined by

$$a_{ij} = \begin{cases} \sigma(e_{ij}), & \text{if } v_i \text{ is adjacent to } v_j, \\ 0, & \text{otherwise.} \end{cases}$$

The adjacency spectrum of Σ consists of the eigenvalues of its adjacency matrix A_Σ, including multiplicities.

Let $d_i = d_G(v_i)$ be the degree of vertex v_i in G and let D_G be the $n \times n$ diagonal matrix with diagonal entries $d_1, d_2, \ldots, d_n$. The Laplacian matrix L_Σ of a signed graph $\Sigma = (G, \sigma)$ is defined by $L_\Sigma = D_G - A_\Sigma$.

Similarly the Laplacian spectrum of Σ consists of the eigenvalues of its Laplacian matrix L_Σ, including multiplicities.

Normalised Laplacian matrices

Taking the definition provided by Chung[1] to signed graphs, the normalised Laplacian matrix $\mathcal{L}_\Sigma$ of a signed graph $\Sigma = (G, \sigma)$ is defined by $\mathcal{L}_\Sigma = D_G^{-\frac{1}{2}} L_\Sigma D_G^{-\frac{1}{2}}$ with the convention that if the degree of vertex v_i in G is zero then $(d_i)^{-\frac{1}{2}} = 0$. Formally

$$(\mathcal{L}_\Sigma)_{ij} = \begin{cases} 1, & \text{if } i = j \text{ and } d_i \neq 0, \\ -\sigma(e_{ij}) \frac{1}{\sqrt{d_i d_j}}, & \text{if } i \neq j \text{ and } v_i \text{ is adjacent to } v_j, \\ 0, & \text{otherwise.} \end{cases}$$

As you would expect, the normalised Laplacian spectrum of Σ consists of the eigenvalues of its normalised Laplacian matrix $\mathcal{L}_\Sigma$, including multiplicities.

[1] F. Chung. Spectral Graph Theory. American Mathematical Society, Providence, Rhode Island. 1997.

Net-degree and co-regularity

The net-degree $d^{\pm}_{\Sigma}(v)$ of a vertex v in a signed graph Σ is defined as $d^{\pm}_{\Sigma}(v) = d^{+}_{\Sigma}(v) - d^{-}_{\Sigma}(v)$ where $d^{+}_{\Sigma}(v)$ and $d^{-}_{\Sigma}(v)$ denote respectively the number of positive edges and the number of negative edges incident with v.

The notion of co-regularity was brought forth in 2015 by Hameed et al.[2] There they defined a signed graph $\Sigma = (G, \sigma)$ to be co-regular if the underlying graph G is r-regular for some positive integer r and the signed graph Σ is net-regular with net-degree k for some integer k. In this case the co-regularity pair is also defined to be the ordered pair (r, k).

[2] S. Hameed, V. Paul, K.A. Germina. On co-regular signed graphs. Australasian Journal of Combinatorics; 62(1); 8-17. 2015.

Cospectrality and spectral determination

For an $n \times n$ matrix M we denote the characteristic polynomial $\det(xI_n - m)$ of M by $P(M; x)$.

For a signed graph Σ we say that $P(X_\Sigma; x)$ is the X-characteristic polynomial of Σ for $X \in \{A, L, \mathcal{L}\}$.

Two signed graphs Σ and Γ are said to be X-cospectral if they have the same X-spectrum. A pair of graphs are said to be cospectral mates if they have the same spectrum but are not isomorphic. We are interested in constructing such pairs.

Here, if two or more X-cospectral graphs are not isomorphic then we say that they are X-nonisomorphic-cospectral or XNICS for short.

A signed graph Σ is said to be determined by its X-spectrum if every signed graph Γ that is X-cospectral with Σ is also isomorphic to Σ.

A pair of co-regular NICS signed graphs

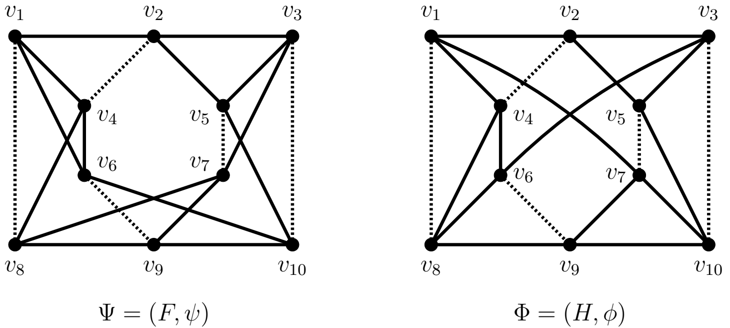

Figure: A pair of (4,2)-co-regular nonisomorphic cospectral signed graphs Ψ and Φ. The edge connecting v_1 and v_4 is in three triangles in Ψ but only two triangles in Φ. Here and throughout, solid lines represent positive edges and dashed lines represent negative edges.

Switching equivalence

An important concept in signed graph theory is the operation of switching. If U is a subset of the vertices V of Σ then the switched signed graph Σ^U is obtained from Σ by reversing the signs of the edges in the cut $[U, V \setminus U]$. The signed graphs Σ and Σ^U are said to be switching equivalent, written $\Sigma \sim \Sigma^U$, and the same is said of their signatures.

It is well known that two signed graphs $\Sigma_1 = (G_1, \sigma_1)$ and $\Sigma_2 = (G_2, \sigma_2)$ on the same vertex set are switching equivalent if and only if their adjacency matrices satisfy $A_{\Sigma_2} = S^{-1} A_{\Sigma_1} S$ for some diagonal matrix S, known as the switching matrix, whose diagonal only has entries of either $+1$ or -1.

Switching a signed graph does not change its spectrum because conjugating a matrix does not change its spectrum. Therefore any two switching equivalent signed graphs share the same characteristic polynomial, spectrum, and eigenvalues.

NS and NNS join definitions

The join of two ordinary graphs G_1 and G_2 is the graph obtained by joining each vertex of G_1 with every vertex of G_2. This is denoted by $G_1 \vee G_2$. For signed graphs, we shall mainly focus on the case where each new introduced edge is made to be positive and we will call this the positive join of Σ_1 and Σ_2 and write it as $\Sigma_1 \vee_+ \Sigma_2$.

Definition (NS join. Adapted from Lu et al.[3])

Let $\Sigma_1 = (G_1, \sigma_1)$ and $\Sigma_2 = (G_2, \sigma_2)$ be two vertex-disjoint signed graphs with n_1 and n_2 vertices respectively. Let $V(\Sigma_1) = \{v_1, v_2, \ldots, v_{n_1}\}$. Then the neighbours splitting (NS) join of Σ_1 and Σ_2 denoted by $\Sigma_1 \veebar \Sigma_2$ is the graph obtained by adding a new vertex v'_i for each $v_i \in V(\Sigma_1)$ to the positive join $\Sigma_1 \vee_+ \Sigma_2$ and connecting v'_i to v_j with a positive edge if and only if v_i is adjacent to v_j in Σ_1.

Definition (NNS join. Adapted from Hamud and Berman[4].)

In the same setting, the non-neighbours splitting (NNS) join of Σ_1 and Σ_2 denoted by $\Sigma_1 \overline{\wedge} \Sigma_2$ is the graph obtained by adding a new vertex v'_i for each $v_i \in V(\Sigma_1)$ to the positive join $\Sigma_1 \vee_+ \Sigma_2$ and connecting v'_i to v_j with a positive edge if and only if $i \neq j$ and v_i is not adjacent to v_j in Σ_1.

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[3] Z. Lu, X. Ma, M. Zhang. Spectra of graph operations based on splitting graph. Journal of Applied Analysis and Computation; 13(1); 133-155. 2023.

[4] S. Hamud, A. Berman. New constructions of nonregular cospectral graphs. Special Matrices; 12(1); 20230109. 2024.

NS and NNS join examples

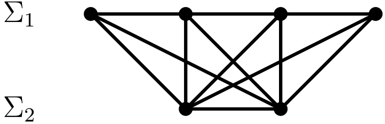
Σ_1 

NS and NNS join examples

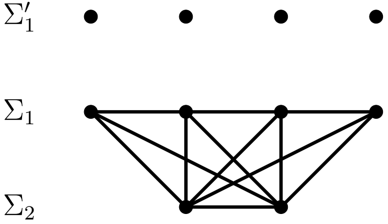
Σ_1 

Σ_2 

NS and NNS join examples



NS and NNS join examples

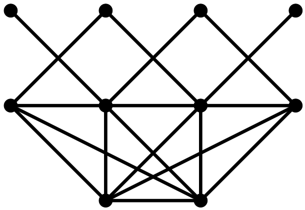


NS and NNS join examples

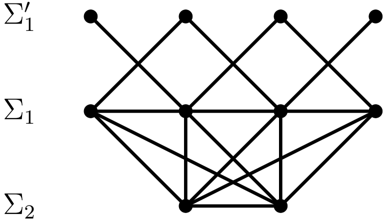
Σ_1'

Σ_1

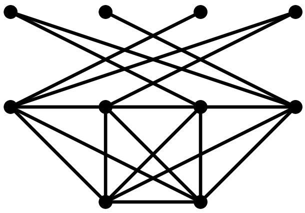
Σ_2



NS and NNS join examples



$P_4 \,\underline{\vee}\, P_2$



$P_4 \,\overline{\wedge}\, P_2$

Preliminaries

Eigenvalues

The adjacency eigenvalues of a graph are denoted by λ. The Laplacian eigenvalues of a graph are denoted by μ. The normalised Laplacian eigenvalues of a graph are denoted by δ, however these will not be used since they introduce some difficulties when consider 0-net-regular graphs.

Remark

Let us write the adjacency eigenvalues of a (r,k)-co-regular signed graph $\Sigma = (G,\sigma)$ on n vertices by first identifying $\lambda_1(\Sigma) = k$ and then ordering the remaining eigenvalues such that $\lambda_2(\Sigma) \geq \lambda_3(\Sigma) \geq \cdots \geq \lambda_n(\Sigma)$. For the adjacency eigenvalues of the underlying graph G we can simply have $\lambda_1(G) = r \geq \lambda_2(G) \geq \lambda_3(G) \geq \cdots \geq \lambda_n(G)$.

Remark

Let us write the Laplacian eigenvalues of a (r,k)-co-regular signed graph $\Sigma = (G,\sigma)$ on n vertices by first identifying $\mu_1(\Sigma) = r - k$ and then ordering the remaining eigenvalues such that $\mu_2(\Sigma) \leq \mu_3(\Sigma) \leq \cdots \leq \mu_n(\Sigma)$. For the Laplacian eigenvalues of the underlying graph G we can simply have $\mu_1(G) = 0 \leq \mu_2(G) \leq \mu_3(G) \leq \cdots \leq \mu_n(G)$.

The coronal χ of a matrix

The graph invariant known as the coronal was introduced in 2011 by McLeman and McNicholas[5]. It is as follows.

Definition

Let M be an $n \times n$ matrix. The coronal $\chi_M(x)$ of M is defined to be the sum of the entries of the inverse of the characteristic matrix of M. This can be calculated as

$$\chi_M(x) = \mathbf{1}_n^T (xI_n - M)^{-1} \mathbf{1}_n.$$

We also have the following useful lemma which can be applied, for instance, when considering the adjacency matrix of a signed graph that is net-regular with net-degree k.

Lemma

Let M be an $n \times n$ matrix with all row sums equal to k. Then

$$\chi_M(x) = \frac{n}{x-k}.$$

[5] C. McLeman, E. McNicholas. Spectra of coronae. Linear Algebra and its Applications; 435; 998-1007. 2011.

Schur's complement formula

The following result is known as Schur's complement formula and can be found in several textbooks.

Lemma

Let $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ be a block matrix such that A and D are square matrices. Then

(a) If A is invertible then $\det(M) = \det(A)\det\left(D - CA^{-1}B\right)$.

(b) If D is invertible then $\det(M) = \det(D)\det\left(A - BD^{-1}C\right)$.

This formula can be extended to simplify finding the determinants of block matrices structured in different ways. It will be helpful to have this specific result presented by Hamud and Berman in the same paper as was referenced before.

Lemma

Let M be a block matrix

$$M = \begin{pmatrix} A & B & J_{n_1 \times n_2} \\ B & C & O_{n_1 \times n_2} \\ J_{n_2 \times n_1} & O_{n_2 \times n_1} & D \end{pmatrix}$$

where A, B, and C are square matrices of order n_1 and D is a square matrix of order n_2. Then for the characteristic determinant of M we have

$$\det\left(xI_{2n_1+n_2} - M\right) = \det\left(xI_{n_2} - D\right)\det\begin{pmatrix} xI_{n_1} - A - \chi_D(x)J_{n_1} & -B \\ -B & xI_{n_1} - C \end{pmatrix}.$$

An extended complement formula

In fact this can be generalised a little further by introducing a scalar component along with the all-one matrices $J_{n_1 \times n_2}$ and $J_{n_2 \times n_1}$.

Lemma

Let M be a block matrix

$$M = \begin{pmatrix} A & B & \omega J_{n_1 \times n_2} \\ B & C & O_{n_1 \times n_2} \\ \omega J_{n_2 \times n_1} & O_{n_2 \times n_1} & D \end{pmatrix}$$

where A, B, and C are square matrices of order n_1 and D is a square matrix of order n_2 and ω is any real number. Then for the characteristic determinant of M we have

$$\det\left(xI_{2n_1+n_2} - M\right) = \det\left(xI_{n_2} - D\right) \det \begin{pmatrix} xI_{n_1} - A - \omega^2 \chi_D(x) J_{n_1} & -B \\ -B & xI_{n_1} - C \end{pmatrix}.$$

It is clear to see that the lemma on the previous slide is the same as the lemma on this slide in the case where $\omega = 1$.

Proof for the previous lemma

Proof.

The characteristic matrix of M is

$$xI_{2n_1+n_2} - M = \begin{pmatrix} xI_{n_1} - A & -B & -\omega J_{n_1\times n_2} \\ -B & xI_{n_1} - C & O_{n_1\times n_2} \\ -\omega J_{n_2\times n_1} & O_{n_2\times n_1} & xI_{n_2} - D \end{pmatrix}.$$

Hence by the invertibility of $xI_{n_2} - D$ and Schur's complement formula we have the determinant of the characteristic matrix of M as

$$\begin{aligned}
&= \det\left(xI_{n_2} - D\right) \det\left(\begin{pmatrix} xI_{n_1} - A & -B \\ -B & xI_{n_1} - C \end{pmatrix}\right. \\
&\qquad\left. - \begin{pmatrix} -\omega J_{n_1\times n_2} \\ O_{n_1\times n_2} \end{pmatrix} (xI_{n_2} - D)^{-1} \begin{pmatrix} -\omega J_{n_2\times n_1} & O_{n_2\times n_1} \end{pmatrix}\right) \\
&= \det\left(xI_{n_2} - D\right) \det\left(\begin{pmatrix} xI_{n_1} - A & -B \\ -B & xI_{n_1} - C \end{pmatrix}\right. \\
&\qquad\left. -\omega^2 \begin{pmatrix} \mathbf{1}_{n_1}\mathbf{1}_{n_2}^T \\ O_{n_1\times n_2} \end{pmatrix} (xI_{n_2} - D)^{-1} \begin{pmatrix} \mathbf{1}_{n_2}\mathbf{1}_{n_1}^T & O_{n_2\times n_1} \end{pmatrix}\right) \\
&= \det\left(xI_{n_2} - D\right) \det\begin{pmatrix} xI_{n_1} - A - \omega^2 \chi_D(x) J_{n_1} & -B \\ -B & xI_{n_1} - C \end{pmatrix}.
\end{aligned}$$

□

The adjugate of a matrix and a useful result

The adjugate of a matrix is defined so that the product of a matrix A with its adjugate $\operatorname{adj}(A)$ yields a diagonal matrix whose diagonal entries are the determinant $\det(A)$. Consequently, when A is invertible, we have that $\operatorname{adj}(A) = \det(A)A^{-1}$.

This final result can be found in the book by Cvetković et al.[6] and will also be utilised in upcoming proofs.

Lemma

If A is an $n \times n$ real matrix and a is a real number then

$$\det(A + aJ_n) = \det(A) + a\mathbf{1}_n^T \operatorname{adj}(A)\mathbf{1}_n.$$

[6] D.M. Cvetković, P. Rowlinson, S.K. Simić. An Introduction to the Theory of Graph Spectra. London Mathematical Society Student Texts, Cambridge University, London. 2010.

Characteristic polynomials

Adjacency characteristic polynomial

Theorem

Let $\Sigma_1 = (G_1, \sigma_1)$ be a signed graph on n_1 vertices and let $\Sigma_2 = (G_2, \sigma_2)$ be a signed graph on n_2 vertices. Then

(a) The adjacency characteristic polynomial of the NS join $\Sigma_1 \veebar \Sigma_2$ is

$$P(A_{\Sigma_1 \veebar \Sigma_2}; x) = x^{n_1} P\big(A_{\Sigma_2}; x\big) \det\left(xI_{n_1} - A_{\Sigma_1} - \frac{1}{x} A_{G_1}^2\right) \left[1 - \chi_{A_{\Sigma_2}}(x) \chi_{A_{\Sigma_1} + \frac{1}{x} A_{G_1}^2}(x)\right].$$

(b) The adjacency characteristic polynomial of the NNS join $\Sigma_1 \barwedge \Sigma_2$ is

$$P(A_{\Sigma_1 \barwedge \Sigma_2}; x) = x^{n_1} P\big(A_{\Sigma_2}; x\big) \det\left(xI_{n_1} - A_{\Sigma_1} - \frac{1}{x} A_{\bar{G}_1}^2\right) \left[1 - \chi_{A_{\Sigma_2}}(x) \chi_{A_{\Sigma_1} + \frac{1}{x} A_{\bar{G}_1}^2}(x)\right].$$

Proof for the adjacency characteristic polynomial

Proof.

We shall provide a proof for the NNS join. By labelling the vertices of $\Sigma_1 \overline{\wedge} \Sigma_2$ in a suitable order we obtain the adjacency matrix

$$A_{\Sigma_1 \overline{\wedge} \Sigma_2} = \begin{pmatrix} A_{\Sigma_1} & A_{\bar{G}_1} & J_{n_1 \times n_2} \\ A_{\bar{G}_1} & O_{n_1} & O_{n_1 \times n_2} \\ J_{n_2 \times n_1} & O_{n_2 \times n_1} & A_{\Sigma_2} \end{pmatrix}.$$

Thus by applying the lemma obtained by extending Schur's complement formula we get the adjacency characteristic polynomial

$$P(A_{\Sigma_1 \overline{\wedge} \Sigma_2}; x) = \det\big(xI_{n_2} - A_{\Sigma_2}\big) \det \begin{pmatrix} xI_{n_1} - A_{\Sigma_1} - \chi_{\Sigma_2}(x) J_{n_1} & -A_{\bar{G}_1} \\ -A_{\bar{G}_1} & xI_{n_1} \end{pmatrix}.$$

Now we can apply Schur's complement formula and the result we have pertaining to the adjugate of a matrix to acquire

$$\begin{aligned}
&= \det\left(xI_{n_2} - A_{\Sigma_2}\right) \det\left(xI_{n_1}\right) \det\left(xI_{n_1} - A_{\Sigma_1} - \chi_{A_{\Sigma_2}}(x) J_{n_1} - A_{\bar{G}_1} (xI_{n_1})^{-1} A_{\bar{G}_1}\right) \\
&= x^{n_1} P\big(A_{\Sigma_2}; x\big) \det\left(xI_{n_1} - A_{\Sigma_1} - \frac{1}{x} A^2_{\bar{G}_1} - \chi_{A_{\Sigma_2}}(x) J_{n_1}\right) \\
&= x^{n_1} P\big(A_{\Sigma_2}; x\big) \left[\det\left(xI_{n_1} - A_{\Sigma_1} - \frac{1}{x} A^2_{\bar{G}_1}\right) - \chi_{A_{\Sigma_2}}(x) \mathbf{1}^T_{n_1} \operatorname{adj}\left(xI_{n_1} - A_{\Sigma_1} - \frac{1}{x} A^2_{\bar{G}_1}\right) \mathbf{1}_{n_1}\right] \\
&= x^{n_1} P\big(A_{\Sigma_2}; x\big) \det\left(xI_{n_1} - A_{\Sigma_1} - \frac{1}{x} A^2_{\bar{G}_1}\right) \left[1 - \chi_{A_{\Sigma_2}}(x) \mathbf{1}^T_{n_1} \left(xI_{n_1} - A_{\Sigma_1} - \frac{1}{x} A^2_{\bar{G}_1}\right)^{-1} \mathbf{1}_{n_1}\right].
\end{aligned}$$

Laplacian characteristic polynomial

Theorem

Let $\Sigma_1 = (G_1, \sigma_1)$ be a signed graph on n_1 vertices and let $\Sigma_2 = (G_2, \sigma_2)$ be a signed graph on n_2 vertices. Then

(a) The Laplacian characteristic polynomial of the NS join $\Sigma_1 \veebar \Sigma_2$ is $P(L_{\Sigma_1 \veebar \Sigma_2}; x) =$

$$\det\left((x - n_1)I_{n_2} - L_{\Sigma_2}\right) \det\left(xI_{n_1} - D_{G_1}\right)$$
$$\det\left((x - n_2)I_{n_1} - L_{\Sigma_1} - D_{G_1} - A_{G_1}\left(xI_{n_1} - D_{G_1}\right)^{-1} A_{G_1}\right)$$
$$\left[1 - \chi_{L_{\Sigma_2}}(x - n_1)\chi_{L_{\Sigma_1} + D_{G_1} + A_{G_1}(xI_{n_1} - D_{G_1})^{-1}A_{G_1}}(x - n_2)\right].$$

(b) The Laplacian characteristic polynomial of the NNS join $\Sigma_1 \barwedge \Sigma_2$ is $P(L_{\Sigma_1 \barwedge \Sigma_2}; x) =$

$$\det\left((x - n_1)I_{n_2} - L_{\Sigma_2}\right) \det\left((x - n_1 + 1)I_{n_1} + D_{G_1}\right)$$
$$\det\left((x - n_1 - n_2 + 1)I_{n_1} - L_{\Sigma_1} + D_{G_1} - A_{\bar{G}_1}\left((x - n_1 + 1)I_{n_1} + D_{G_1}\right)^{-1} A_{\bar{G}_1}\right)$$
$$\left[1 - \chi_{L_{\Sigma_2}}(x - n_1)\chi_{L_{\Sigma_1} - D_{G_1} + A_{\bar{G}_1}((x - n_1 + 1)I_{n_1} + D_{G_1})^{-1}A_{\bar{G}_1}}(x - n_1 - n_2 + 1)\right].$$

Normalised Laplacian characteristic polynomial (i)

In order to obtain the characteristic polynomials of the normalised Laplacian matrices we shall impose the condition that the signed graphs must have regular underlying graphs. Without this we are unable to make proper use of a previous lemma and the computation becomes much more involved. Fortunately this restriction will have no bearing on the results obtained later on.

Let $\Sigma_1 = (G_1, \sigma_1)$ be a signed graph on n_1 vertices with G_1 being r_1 regular and let $\Sigma_2 = (G_2, \sigma_2)$ be a signed graph on n_2 vertices with G_2 being r_2 regular. If the underlying graph $G_1 = K_{n_1}$, the complete graph on n_1 vertices, then its complement will be the edgeless graph on n_1 vertices. Hence the adjacency matrix of its complement will be $A_{\bar{G}_1} = O_{n_1}$.

This is important to note because as we follow the steps for making the NNS join we find that each newly introduced vertex will have no 'non-neighbour' to connect to and so will be isolated, i.e. have degree equal to zero. This will require us to make the distinction between the cases where $G_1 = K_{n_1}$ and $G_1 \neq K_{n_1}$ when calculating the normalised Laplacian characteristic polynomial of the NNS join.

Normalised Laplacian characteristic polynomial (ii)

Theorem

Let $\Sigma_1 = (G_1, \sigma_1)$ be a signed graph on n_1 vertices with G_1 being r_1 regular and let $\Sigma_2 = (G_2, \sigma_2)$ be a signed graph on n_2 vertices with G_2 being r_2 regular. Then

(a) Let $\alpha = \frac{1}{n_2+2r_1}$, $\beta = \frac{1}{r_1}$, and $\gamma = \frac{1}{n_1+r_2}$. The normalised Laplacian characteristic polynomial of the NS join $\Sigma_1 \veebar \Sigma_2$ is

$$P(\mathcal{L}_{\Sigma_1 \veebar \Sigma_2}; x) = (x-1)^{n_1} \det\left((x-1)I_{n_2} + \gamma A_{\Sigma_2}\right)$$

$$\det\left((x-1)I_{n_1} + \alpha A_{\Sigma_1} - \frac{\alpha\beta}{x-1} A_{G_1}^2\right)$$

$$\left[1 - \alpha\gamma \chi_{\left(-\gamma A_{\Sigma_2}\right)}(x-1)\chi_{\left(-\alpha A_{\Sigma_1} + \frac{\alpha\beta}{x-1} A_{G_1}^2\right)}(x-1)\right].$$

Normalised Laplacian characteristic polynomial (iii)

Theorem

(b)(i) Let $\alpha = \frac{1}{n_1+n_2-1}$, $\beta = \frac{1}{n_1-r_1-1}$, *and* $\gamma = \frac{1}{n_1+r_2}$. *The normalised Laplacian characteristic polynomial of the NNS join* $\Sigma_1 \overline{\wedge} \Sigma_2$ *when* $G_1 \neq K_{n_1}$ *is*

$$P(\mathcal{L}_{\Sigma_1 \overline{\wedge} \Sigma_2}; x) = (x-1)^{n_1} \det\left((x-1)I_{n_2} + \gamma A_{\Sigma_2}\right) \det\left((x-1)I_{n_1} + \alpha A_{\Sigma_1} - \frac{\alpha\beta}{x-1} A^2_{\bar{G}_1}\right) \left[1 - \alpha\gamma \chi_{\left(-\gamma A_{\Sigma_2}\right)}(x-1) \chi_{\left(-\alpha A_{\Sigma_1} + \frac{\alpha\beta}{x-1} A^2_{\bar{G}_1}\right)}(x-1)\right].$$

(b)(ii) Let $\alpha = \frac{1}{n_1+n_2-1}$ *and* $\gamma = \frac{1}{n_1+r_2}$. *The normalised Laplacian characteristic polynomial of the NNS join* $\Sigma_1 \overline{\wedge} \Sigma_2$ *when* $G_1 = K_{n_1}$ *is*

$$P(\mathcal{L}_{\Sigma_1 \overline{\wedge} \Sigma_2}; x) = x^{n_1} \det\left((x-1)I_{n_2} + \gamma A_{\Sigma_2}\right) \det\left((x-1)I_{n_1} + \alpha A_{\Sigma_1}\right) \left[1 - \alpha\gamma \chi_{\left(-\gamma A_{\Sigma_2}\right)}(x-1) \chi_{\left(-\alpha A_{\Sigma_1}\right)}(x-1)\right].$$

Constructing nonregular NICS pairs

Now that we have computed the various characteristic polynomials we can use the results to construct nonregular nonisomorphic cospectral pairs of signed graphs.

Corollary

Let $\Psi = (F, \psi)$ and $\Phi = (H, \phi)$ both be (r, k)-co-regular of the same valencies and also be nonisomorphic cospectral signed graphs on the same set of vertices. Then for every signed graph $\Sigma = (G, \sigma)$

(a) $\Sigma \veebar \Psi$ *and* $\Sigma \veebar \Phi$ *are* $\{A, L, \mathcal{L}\}$*NICS.*

(b) $\Sigma \,\overline{\wedge}\, \Psi$ *and* $\Sigma \,\overline{\wedge}\, \Phi$ *are* $\{A, L, \mathcal{L}\}$*NICS.*

Proof.

We shall prove (a). The NS join graphs $\Sigma \veebar \Psi$ and $\Sigma \veebar \Phi$ are nonisomorphic since Ψ and Φ are nonisomorphic. By Theorems... we know that

$$P(A_{\Sigma \veebar \Psi}) = P(A_{\Sigma \veebar \Phi}), \quad P(L_{\Sigma \veebar \Psi}) = P(L_{\Sigma \veebar \Phi}), \quad P(\mathcal{L}_{\Sigma \veebar \Psi}) = P(\mathcal{L}_{\Sigma \veebar \Phi}).$$

This is because the pairs of adjacency matrices A_Ψ and A_Φ, Laplacian matrices L_Ψ and L_Φ, and normalised Laplacian matrices $\mathcal{L}_\Psi$ and $\mathcal{L}_\Phi$ all have the same coronal as their relevant partner by the shared (r, k)-co-regularity of Ψ and Φ. By the cospectrality of Ψ and Φ they also have the same characteristic polynomials and this completes the proof of (a). The proof of (b) is similar. $\square$

Constructing more nonregular NICS pairs

We have a way of constructing more such pairs by simply switching the roles played by the graph Σ and the graphs Ψ and Φ.

Corollary

Let $\Psi = (F, \psi)$ and $\Phi = (H, \phi)$ both be (r, k)-co-regular of the same valencies and also be nonisomorphic cospectral signed graphs on the same set of vertices. Then for every signed graph $\Sigma = (G, \sigma)$

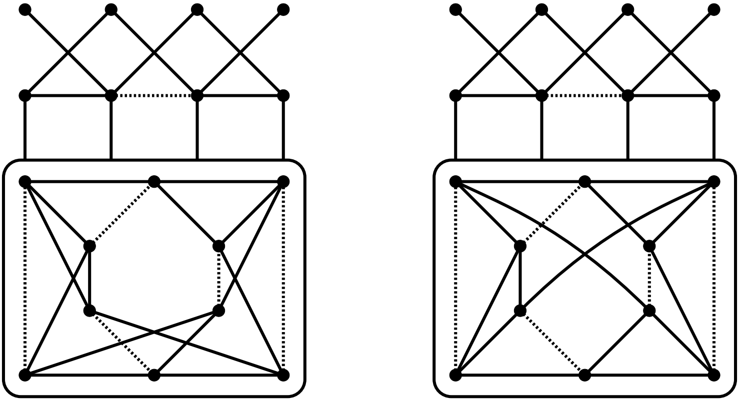
(a) $\Psi \veebar \Sigma$ *and* $\Phi \veebar \Sigma$ *are* $\{A, L, \mathcal{L}\}$*NICS.*

(b) $\Psi \barwedge \Sigma$ *and* $\Phi \barwedge \Sigma$ *are* $\{A, L, \mathcal{L}\}$*NICS.*

Proof.

The proof is very similar to that of the previous corollary. □

A signed nonregular NICS pair



$\Sigma \veebar \Psi$ $\qquad$ $\Sigma \veebar \Phi$

Figure: These are the graphs $\Sigma \veebar \Psi$ and $\Sigma \veebar \Phi$ where $\Sigma = (P_4, \sigma)$ and Ψ and Φ are taken from Figure 1. Together they form a pair of nonregular nonisomorphic cospectral signed graphs. A line between a single vertex and a box surrounding a set of vertices represents the multiple edges connecting the single vertex to every vertex within the box.

Spectra

Adjacency spectra of NS join

Let $\Sigma_1 = (G_1, \sigma_1)$ be an (r_1, k_1)-co-regular graph on n_1 vertices and let $\Sigma_2 = (G_2, \sigma_2)$ be an (r_2, k_2)-co-regular graph on n_2 vertices. Then the adjacency spectrum of $\Sigma_1 \veebar \Sigma_2$ consists of

(i) $\lambda_j(\Sigma_2)$ for each $j = 2, 3, \ldots, n_2$,

(ii) the two roots of the equation

$$x^2 - \lambda_i(\Sigma_1)x - \lambda_i^2(G_1) = 0$$

for each $i = 2, 3, \ldots, n_1$,

(iii) the three roots of the equation

$$x^3 - \left(k_1 + k_2\right)x^2 - \left(r_1^2 - k_1k_2 + n_1n_2\right)x + k_2r_1^2 = 0.$$

Proof for the adjacency spectra of NS join

Proof.

We have previously computed the adjacency characteristic polynomial of $\Sigma_1 \veebar \Sigma_2$ to be

$$P(A_{\Sigma_1 \veebar \Sigma_2}; x) = x^{n_1} P\left(A_{\Sigma_2}; x\right) \det\left(x I_{n_1} - A_{\Sigma_1} - \frac{1}{x} A_{G_1}^2\right)\left[1 - \chi_{A_{\Sigma_2}}(x) \chi_{A_{\Sigma_1} + \frac{1}{x} A_{G_1}^2}(x)\right].$$

Since we are working with co-regular signed graphs we can apply the lemma relating to the coronals of matrices with all row sums equal to the same value to obtain the adjacency characteristic polynomial as

$$= x^{n_1} \prod_{j=1}^{n_2}\left(x - \lambda_j(\Sigma_2)\right) \prod_{i=1}^{n_1}\left(x - \lambda_i(\Sigma_1) - \frac{1}{x}\lambda_i^2(G_1)\right)\left[1 - \frac{n_2}{x - k_2} \cdot \frac{n_1}{x - k_1 - \frac{1}{x} r_1^2}\right]$$

$$= \prod_{j=1}^{n_2}\left(x - \lambda_j(\Sigma_2)\right) \prod_{i=1}^{n_1}\left(x^2 - \lambda_i(\Sigma_1) x - \lambda_i^2(G_1)\right)\left[1 - \frac{n_1 n_2 x}{\left(x - k_2\right)\left(x^2 - k_1 x - r_1^2\right)}\right]$$

$$= \prod_{j=1}^{n_2}\left(x - \lambda_j(\Sigma_2)\right) \prod_{i=1}^{n_1}\left(x^2 - \lambda_i(\Sigma_1) x - \lambda_i^2(G_1)\right)\left[\frac{\left(x - k_2\right)\left(x^2 - k_1 x - r_1^2\right) - n_1 n_2 x}{\left(x - k_2\right)\left(x^2 - k_1 x - r_1^2\right)}\right]$$

$$= \prod_{j=1}^{n_2}\left(x - \lambda_j(\Sigma_2)\right) \prod_{i=1}^{n_1}\left(x^2 - \lambda_i(\Sigma_1) x - \lambda_i^2(G_1)\right)$$

Laplacian spectra of NS join

Let $\Sigma_1 = (G_1, \sigma_1)$ be an (r_1, k_1)-co-regular graph on n_1 vertices and let $\Sigma_2 = (G_2, \sigma_2)$ be an (r_2, k_2)-co-regular graph on n_2 vertices. Then the Laplacian spectrum of $\Sigma_1 \veebar \Sigma_2$ consists of

(i) $n_1 + \mu_j(\Sigma_2)$ for each $j = 2, 3, \ldots, n_2$,

(ii) the two roots of the equation

$$x^2 - \big(n_2 + 2r_1 + \mu_i(\Sigma_1)\big)x + r_1\mu_i(\Sigma_1) - \mu_i^2(G_1) + 2r_1\mu_i(G_1) + n_2r_1 = 0$$

for each $i = 2, 3, \ldots, n_1$,

(iii) the three roots of the equation

$$x^3 + \big(k_1 + k_2 - n_1 - n_2 - 3r_1 - r_2\big)x^2 + \big(r_1^2 + k_1k_2 - k_1n_1 - k_1r_1 - k_1r_2 - k_2n_2 - 3k_2r_1 + 3n_1r_1 + n_2r_1 + n_2r_2 + 3r_1r_2\big)x + k_2r_1^2 - n_1r_1^2 - r_1^2r_2 - k_1k_2r_1 + k_1r_1r_2 + k_2n_2r_1 - n_2r_1r_2 = 0.$$

Normalised Laplacian spectra of NS join

Let $\Sigma_1 = (G_1, \sigma_1)$ be an (r_1, k_1)-co-regular graph on n_1 vertices and let $\Sigma_2 = (G_2, \sigma_2)$ be an (r_2, k_2)-co-regular graph on n_2 vertices. Let $\alpha = \frac{1}{n_1+2r_1}$, $\beta = \frac{1}{r_1}$, and $\gamma = \frac{1}{n_1+r_2}$. Then the normalised Laplacian spectrum of $\Sigma_1 \veebar \Sigma_2$ consists of

(i) $1 - \gamma\lambda_j(\Sigma_2)$ for each $j = 2, 3, \ldots, n_2$,

(ii) the two roots of the equation

$$x^2 - \big(2 - \alpha\lambda_i(\Sigma_1)\big)x - \alpha\lambda_i(\Sigma_1) - \alpha\beta\lambda_i^2(G_1) + 1 = 0$$

for each $i = 2, 3, \ldots, n_1$,

(iii) the three roots of the equation

$$x^3 + \big(\alpha k_1 + \gamma k_2 - 3\big)x^2 - \big(2\alpha k_1 + 2\gamma k_2 + \alpha\beta r_1^2 - \alpha\gamma k_1 k_2 + \alpha\gamma n_1 n_2 - 3\big)x$$
$$+\alpha k_1 + \gamma k_2 + \alpha\beta r_1^2 - \alpha\gamma k_1 k_2 + \alpha\gamma n_1 n_2 - \alpha\beta\gamma k_2 r_1^2 - 1 = 0.$$

Adjacency spectra of NNS join

Let $\Sigma_1 = (G_1, \sigma_1)$ be an (r_1, k_1)-co-regular graph on n_1 vertices and let $\Sigma_2 = (G_2, \sigma_2)$ be an (r_2, k_2)-co-regular graph on n_2 vertices. Then the adjacency spectrum of $\Sigma_1 \overline{\wedge} \Sigma_2$ consists of

(i) $\lambda_j(\Sigma_2)$ for each $j = 2, 3, \dots, n_2$,

(ii) the two roots of the equation

$$x^2 - \lambda_i(\Sigma_1)x - \lambda_i^2(G_1) - 2\lambda_i(G_1) - 1 = 0$$

for each $i = 2, 3, \dots, n_1$,

(iii) the three roots of the equation

$$x^3 - \big(k_1 + k_2\big)x^2 + \big(k_1k_2 - n_1n_2 - (n_1 - r_1 - 1)^2\big)x + k_2(n_1 - r_1 - 1)^2 = 0.$$

Laplacian spectra of NNS join

Let $\Sigma_1 = (G_1, \sigma_1)$ be an (r_1, k_1)-co-regular graph on n_1 vertices and let $\Sigma_2 = (G_2, \sigma_2)$ be an (r_2, k_2)-co-regular graph on n_2 vertices. Then the Laplacian spectrum of $\Sigma_1 \overline{\wedge} \Sigma_2$ consists of

(i) $n_1 + \mu_j(\Sigma_2)$ for each $j = 2, 3, \ldots, n_2$,

(ii) the two roots of the equation

$$x^2 - \big(2n_1 + n_2 - 2r_1 - 2 + \mu_i(\Sigma_1)\big)x + (n_1 - r_1 - 1)\mu_i(\Sigma_1) - \mu_i^2(G_1) \\ + (2r_1 + 2)\mu_i(G_1) + n_1^2 - 2n_1 - n_2 + n_1 n_2 - 2n_1 r_1 - n_2 r_1 = 0.$$

for each $i = 2, 3, \ldots, n_1$,

(iii) the three roots of the equation

$$x^3 + \big(k_1 + k_2 - 3n_1 - n_2 + r_1 - r_2 + 2\big)x^2 + \big(2n_1^2 - r_1^2 + k_1 + 2k_2 - 2n_1 - n_2 \\ - r_1 - 2r_2 + k_1k_2 - 2k_1n_1 + k_1r_1 - k_1r_2 - 2k_2n_1 - k_2n_2 + k_2r_1 + n_1n_2 \\ + 2n_1r_2 - n_2r_1 + n_2r_2 - r_1r_2\big)x + k_1n_1^2 - k_2r_1^2 - n_1^2r_1 + n_1r_1^2 + r_1^2r_2 + k_1k_2 \\ - k_1n_1 - k_1r_2 - k_2n_2 - k_2r_1 + n_1r_1 + n_2r_2 + r_1r_2 - k_1k_2n_1 + k_1k_2r_1 \\ - k_1n_1r_1 + k_1n_1r_2 - k_1r_1r_2 + k_2n_1n_2 + k_2n_1r_1 - k_2n_2r_1 - n_1n_2r_2 \\ - n_1r_1r_2 + n_2r_1r_2 = 0.$$

Normalised Laplacian spectra of NNS join (a)

Since we required distinct cases when calculating the normalised Laplacian characteristic polynomial of the NNS join, we shall require the same for the spectrum.

Let $\Sigma_1 = (G_1, \sigma_1)$ be an (r_1, k_1)-co-regular graph on n_1 vertices and let $\Sigma_2 = (G_2, \sigma_2)$ be an (r_2, k_2)-co-regular graph on n_2 vertices.

(a) Let $\alpha = \frac{1}{n_1+n_2-1}$, $\beta = \frac{1}{n_1-r_1-1}$, and $\gamma = \frac{1}{n_1+r_2}$. Then the normalised Laplacian spectrum of $\Sigma_1 \overline{\wedge} \Sigma_2$ when $G_1 \neq K_{n_1}$ consists of

   (i) $1 - \gamma\lambda_j(\Sigma_2)$ for each $j = 2, 3, \ldots, n_2$,

   (ii) the two roots of the equation

$$x^2 - \big(2 - \alpha\lambda_i(\Sigma_1)\big)x - \alpha\lambda_i(\Sigma_1) - \alpha\beta\lambda_i^2(G_1) - 2\alpha\beta\lambda_i(G_1) - \alpha\beta + 1 = 0.$$

   for each $i = 2, 3, \ldots, n_1$,

   (iii) the three roots of the equation

$$\begin{gathered} x^3 + \big(\alpha k_1 + \gamma k_2 - 3\big)x^2 - \big(2\alpha k_1 + 2\gamma k_2 + \alpha\beta(n_1 - r_1 - 1)^2 - \alpha\gamma k_1 k_2 \\ + \alpha\gamma n_1 n_2 - 3\big)x + \alpha k_1 + \gamma k_2 + \alpha\beta(n_1 - r_1 - 1)^2 - \alpha\gamma k_1 k_2 \\ + \alpha\gamma n_1 n_2 - \alpha\beta\gamma k_2(n_1 - r_1 - 1)^2 - 1 = 0. \end{gathered}$$

Normalised Laplacian spectra of NNS join (b)

Let $\Sigma_1 = (G_1, \sigma_1)$ be an (r_1, k_1)-co-regular graph on n_1 vertices and let $\Sigma_2 = (G_2, \sigma_2)$ be an (r_2, k_2)-co-regular graph on n_2 vertices.

(b) Let $\alpha = \frac{1}{n_1+n_2-1}$ and $\gamma = \frac{1}{n_1+r_2}$. Then the normalised Laplacian spectrum of $\Sigma_1 \overline{\wedge} \Sigma_2$ when $G_1 = K_{n_1}$ consists of

(i) $1 - \gamma\lambda_j(\Sigma_2)$ for each $j = 2, 3, \ldots, n_2$,

(ii) the two roots of the equation

$$x^2 - \big(1 - \alpha\lambda_i(\Sigma_1)\big)x = 0.$$

for each $i = 2, 3, \ldots, n_1$,

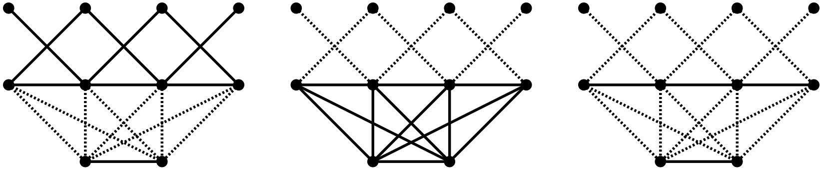
(iii) the three roots of the equation

$$x^3 + \big(\alpha k_1 + \gamma k_2 - 2\big)x^2 - \big(\alpha k_1 + \gamma k_2 - \alpha\gamma k_1 k_2 + \alpha\gamma n_1 n_2 - 1\big)x = 0.$$

Variations

Negative joining edges

Recall the definitions of the NS and NNS joins. Previously we specified that all of the newly introduced edges were to be given positive signature. We could also consider what would happen if some of these edges were instead made to be negative between distinct sets of vertices.



$P_4 \veebar^1 P_2$ $\quad$ $P_4 \veebar^2 P_2$ $\quad$ $P_4 \veebar^3 P_2$

In each of the above cases, the graphs are switching equivalent to the graphs obtained by the NS and NNS joins we have been working with and so share the same characteristic polynomials and spectra. So, while they may not be the most interesting, we can build more pairs of nonregular nonisomorphic cospectral signed graphs.

Splitting S-vertex join

When introducing their version of what we have been calling the NS join, Lu et al. also presented another join which we have not yet looked at. This, along with its 'non-neighbours' form, can be considered for signed graphs in much the same way.

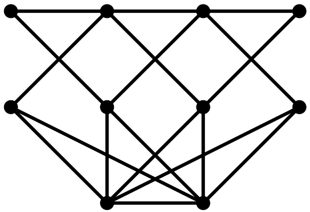
Definition (NS-S join.)

Let $\Sigma_1 = (G_1, \sigma_1)$ and $\Sigma_2 = (G_2, \sigma_2)$ be two vertex-disjoint signed graphs with n_1 and n_2 vertices respectively. Let $V(\Sigma_1) = \{v_1, v_2, \ldots, v_{n_1}\}$. Then the neighbours splitting S (NS-S) join of Σ_1 and Σ_2 denoted by $\Sigma_1 \veebar^S \Sigma_2$ is the graph obtained by adding a new vertex v_i' for each $v_i \in V(\Sigma_1)$, making the positive join $\Sigma_1' \vee_+ \Sigma_2$ and connecting v_i' to v_j with a positive edge if and only if v_i is adjacent to v_j in Σ_1.

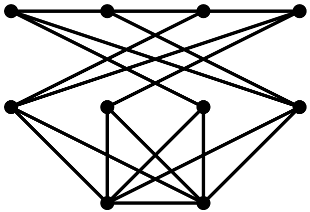
Definition (NNS-S join.)

In the same setting, the non-neighbours splitting S (NNS-S) join of Σ_1 and Σ_2 denoted by $\Sigma_1 \overline{\wedge}^S \Sigma_2$ is the graph obtained by adding a new vertex v_i' for each $v_i \in V(\Sigma_1)$, making the positive join $\Sigma_1' \vee_+ \Sigma_2$ and connecting v_i' to v_j with a positive edge if and only if v_i is not adjacent to v_j in Σ_1.

Splitting join graph examples



$P_4 \veebar^S P_2$



$P_4 \barwedge^S P_2$

Figure: These are the join graphs $P_4 \veebar^S P_2$ and $P_4 \barwedge^S P_2$.

A similar approach, though a little more involved, can be taken to find the characteristic polynomials and spectra of these types of graphs and then we can see that even more pairs of nonregular nonisomorphic cospectral signed graphs can be constructed.

Open problems

- Express the spectra of the normalised Laplacian matrices of the NS and NNS joins in terms of the normalised Laplacian eigenvalues of the component signed graphs.
- Compute the characteristic polynomials of the normalised Laplacian matrices of the NS and NNS joins of any two arbitrary signed graphs.
- Compute the adjacency, Laplacian, and normalised Laplacian spectra of the NS and NNS joins of any two arbitrary signed graphs.
- If possible, define a join operation of two signed graphs Σ_1 and Σ_2 following similar steps to the joins presented here but in a manner such that the resultant compound graphs give rise to the same characteristic polynomials and spectra up to switching equivalence of Σ_1 and Σ_2.

Thank you for listening.

感谢您的聆听。

Хвала што сте саслушали.